



DESIGN AND FABRICATION OF BIPEDAL LOCOMOTION ROBOT



MINI PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Communication barriers faced by individuals with hearing and speech impairments highlight the need for inclusive assistive technologies. This project presents the design and fabrication of a bipedal assistive communication robot equipped with a sign language display and voice interface. Unlike conventional humanoid systems, the robot is developed without arms, focusing on mobility and multimodal communication through visual and auditory outputs. The system aims to bridge interaction gaps by providing an accessible platform capable of both interpreting and conveying information effectively. The robot is designed as a two-legged mechanism to achieve stable and human-like locomotion using lightweight materials and actuator-based motion control. A microcontroller-based system integrates the main functionalities, including text-to-speech conversion for voice output and a display module to present sign language symbols or gestures. Predefined programming enables the robot to process input commands and convert them into synchronized visual and audio communication. The absence of arms reduces mechanical complexity while maintaining efficient and reliable system performance. This system can be applied in public spaces, educational institutions, and healthcare environments to support individuals with hearing and speech impairments. It enhances accessibility by translating spoken information into sign language displays and providing voice assistance for interaction. Furthermore, the robot contributes to the development of inclusive human-robot interaction systems and can serve as a foundation for future advancements in assistive robotic technologies.

Keywords: Assistive robotics, bipedal robot, multimodal communication, accessibility, human–robot interaction, sign language display

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LIST OF ABBREVIATIONS

AI	-	Artificial Intelligence
ADC	-	Analog to Digital Converter
CAN	-	Controller Area Network
DAC	-	Digital to Analog Converter
DC	-	Direct Current
DOF	-	Degree of Freedom
ESP32	-	Espressif 32-bit Microcontroller
FRI	-	Foot Rotation Indicator
GND	-	Ground
I2C	-	Inter-Integrated Circuit
IoT	-	Internet of Things
IR	-	Infrared
LiDAR	-	Light Detection and Ranging
PCB	-	Printed Circuit Board
PIC	-	Peripheral Interface Controller
PID	-	Proportional Integral Derivative

PWM	-	Pulse Width Modulation
RF	-	Radio Frequency
SDA	-	Serial Data Line
SCL	-	Serial Clock Line
SPI	-	Serial Peripheral Interface
UART	-	Universal Asynchronous Receiver Transmitter
VCC	-	Voltage at Common Collector
Wi-Fi	-	Wireless Fidelity
ZMP	-	Zero Moment Point

LIST OF SYMBOLS

V	-	Voltage
A	-	Current
W	-	Power
Ω	-	Resistance
Hz	-	Frequency
°	-	Degree
kg	-	Kilogram
cm	-	Centimeter
mm	-	Millimeter
m	-	Meter
Nm	-	Torque
ms	-	Millisecond
s	-	Second
RPM	-	Revolutions Per Minute
m/s	-	Velocity
m/s ²	-	Acceleration

CHAPTER 1

INTRODUCTION

The development of bipedal locomotion robots has become one of the most significant areas of research in robotics and automation. A bipedal robot is a machine designed to walk and move using two legs, similar to human beings. These robots are capable of balancing, walking, turning, climbing stairs, and performing various tasks in environments designed for humans. The study of bipedal locomotion combines concepts from mechanical engineering, electronics, control systems, artificial intelligence, and biomechanics. The increasing demand for intelligent and adaptive robotic systems has encouraged researchers to develop efficient and stable bipedal robots for industrial, medical, military, and domestic applications.

The design and fabrication of a bipedal locomotion robot involve the integration of mechanical structures, actuators, sensors, controllers, and programming algorithms. The mechanical design focuses on creating a lightweight yet strong structure capable of maintaining balance during movement. Servo motors or actuators are commonly used to imitate the motion of human joints such as hips, knees, and ankles. Sensors like gyroscopes and accelerometers help the robot detect its orientation and maintain stability while walking. The fabrication process includes selecting suitable materials, assembling components, wiring electronic circuits, and implementing control software to coordinate movements accurately.

One of the major challenges in developing a bipedal robot is achieving stable and smooth locomotion. Unlike wheeled robots, bipedal robots must continuously maintain balance while moving on different surfaces. This requires precise coordination between the legs and real-time feedback from sensors. Advanced control techniques such as inverse kinematics, zero moment point

(ZMP), and dynamic balancing are often used to improve walking performance. Researchers aim to develop robots that can adapt to uneven terrains, avoid obstacles, and mimic natural human walking patterns with high efficiency and low power consumption.

Bipedal locomotion robots have a wide range of practical applications in modern society. In healthcare, they can assist elderly or disabled individuals by providing support in rehabilitation and mobility. In disaster management and rescue operations, these robots can access narrow or hazardous areas where humans may face risks. Industrial sectors use humanoid robots for repetitive and hazardous tasks, while educational institutions use them as learning platforms for robotics and artificial intelligence. Additionally, space exploration agencies are exploring the use of bipedal robots for missions in environments where human-like movement is advantageous.

The objective of designing and fabricating a bipedal locomotion robot is to create a reliable and efficient robotic system capable of imitating human walking behaviour. This project provides valuable knowledge in robot mechanics, embedded systems, motion control, and programming. It also enhances problem-solving and innovation skills by addressing challenges related to balance, coordination, and autonomous movement. As technology continues to advance, bipedal robots are expected to play an important role in future automation systems, contributing to improved productivity, safety, and human assistance in various fields.

1.1 CLASSIFICATION OF BIPEDAL ROBOTS

Bipedal robots are classified into different types based on their actuation system, walking mechanism, structural design, control method, and application. Based on actuation, they are classified as fully actuated and underactuated robots. According to the walking mechanism, they are divided into static walking robots and dynamic walking robots. Based on structural design, they are classified as

humanoid and non-humanoid robots. Depending on the control method, bipedal robots may be autonomous or remote-controlled. They are also classified according to their applications such as research, industrial, service, and rescue robots. This classification helps in understanding the design, movement, and operational characteristics of various bipedal robotic systems.

1.1.1 ACTUATION SYSTEM

The classification based on the actuation system depends on how the joints of the robot are controlled and powered. In this category, bipedal robots are mainly classified into fully actuated robots and underactuated robots. Fully actuated robots use motors or actuators at every joint, allowing complete control over movement and balance. These robots can perform accurate and complex motions but require higher power consumption and sophisticated control systems.

Underactuated robots use fewer actuators compared to the total degrees of freedom of the robot. They depend partly on passive dynamics, gravity, and natural motion for locomotion. These robots are lighter, mechanically simpler, and consume less energy. However, controlling balance and movement in underactuated robots is more challenging compared to fully actuated robots.

1.1.2 WALKING SYSTEM

Based on the walking mechanism, bipedal robots are classified into static walking robots and dynamic walking robots. Static walking robots maintain stability at every stage of movement by keeping the center of gravity within the support area formed by the feet. These robots move slowly and carefully, making them suitable for applications where stability is more important than speed.

Dynamic walking robots maintain balance while continuously moving using momentum, sensors, and advanced control algorithms. These robots can walk faster and produce human-like movement patterns. Dynamic walking requires real-time monitoring and adjustment of body posture, making the control system more complex compared to static walking robots.

1.1.3 STRUCTURAL DESIGN

According to structural design, bipedal robots are classified as humanoid robots and non-humanoid robots. Humanoid robots are designed to resemble the human body with features such as a head, torso, arms, and legs. These robots can interact effectively in environments designed for humans and are widely used in healthcare, education, and service industries.

Non-humanoid robots do not completely imitate human appearance and mainly focus on locomotion and balance. Their design is optimized for specific tasks rather than human resemblance. These robots are commonly used in research laboratories, industrial applications, and military operations where efficient movement and functionality are more important than human-like appearance.

1.1.4 CONTROL METHOD

Bipedal robots are also classified based on their control method into autonomous robots and remote-controlled robots. Autonomous robots can sense their surroundings, process information, and make decisions independently using sensors, microcontrollers, and artificial intelligence techniques. These robots can adapt to changes in the environment and perform tasks without direct human control.

Remote-controlled robots depend on external commands provided by an operator through wireless or wired communication systems. The movement and actions of the robot are controlled manually using a controller or computer. These robots are mainly used in experiments, surveillance, and environments where human supervision is necessary.

1.1.5 TERRAIN ADAPTABILITY

Based on terrain adaptability, bipedal robots are classified as indoor robots and outdoor robots. Indoor robots are designed to work on smooth and controlled surfaces such as homes, hospitals, and laboratories. Their movement and balancing systems are optimized for flat environments.

Outdoor robots are designed to move on rough, uneven, or unpredictable terrains. These robots use advanced sensors and balancing mechanisms to walk on stairs, slopes, rocky surfaces, and obstacle-filled environments. Outdoor bipedal robots are mainly used in military, rescue, and exploration applications.

1.1.6 BASED ON APPLICATION

Based on application, bipedal robots are classified as research robots, service robots, industrial robots, and rescue robots. Research robots are mainly developed for studying robotic locomotion, artificial intelligence, balance control, and motion planning. Educational institutions and research laboratories use these robots for experimentation and technological development.

Service robots are designed to assist humans in healthcare, domestic work, and customer service activities. Industrial robots are used in manufacturing and automation processes, while rescue robots are employed in hazardous environments such as disaster zones, mines, and military operations. These robots help perform tasks that may be dangerous or difficult for humans.

1.2 SCOPE AND OBJECTIVE

1.2.1 SCOPE OF THE PROJECT

The scope of the “Design and Fabrication of Bipedal Locomotion Robot” project includes the development of a robotic system capable of walking using two legs while maintaining balance and stability. The project focuses on the mechanical design, fabrication, electronic control, and programming of the robot to achieve basic locomotion such as forward movement, turning, and balancing. It involves the integration of actuators, sensors, microcontrollers, and control algorithms to imitate human walking patterns. The project also covers testing and performance analysis of the robot under different operating conditions.

The scope further extends to studying advanced concepts of robotics such as motion planning, balance control, artificial intelligence, and autonomous navigation. Bipedal robots have wide applications in healthcare, industrial automation, military operations, rescue missions, and research laboratories. This

project provides a foundation for future developments in humanoid robotics and intelligent robotic systems capable of operating in human environments.

1.2.2 OBJECTIVES OF THE PROJECT

- To design and fabricate a stable bipedal locomotion robot capable of walking on two legs.
- To study the principles of human walking and apply them in robotic movement.
- To develop a lightweight and strong mechanical structure for efficient locomotion.
- To integrate actuators, sensors, and microcontrollers for controlled movement and balance.
- To implement walking algorithms for forward, backward, and turning motions.
- To achieve stability and coordination during robot movement.
- To analyze the performance of the robot in terms of balance, speed, and efficiency.
- To gain practical knowledge in robotics, embedded systems, and motion control techniques.
- To explore the applications of bipedal robots in industrial, medical, and rescue fields.

1.3 SUMMARY OF THE WORK

The work carried out in this project mainly focused on the study and development of a bipedal locomotion robot capable of stable two-legged movement. Initially, a detailed study and analysis of the human walking mechanism and the principles of bipedal locomotion were performed to understand balance, coordination, and gait patterns. Based on this analysis, a suitable mechanical structure was selected to achieve stable and efficient walking. The robot body, leg joints, and support frame were designed using lightweight

materials to reduce overall weight while maintaining structural strength and stability. A compact and balanced mechanical assembly was developed to improve locomotion performance and movement efficiency.

For actuation and motion control, appropriate servo motors such as the MG996R Servo Motor were selected and integrated into the robot joints to provide accurate angular movement. Electronic control circuits were designed and implemented for the proper operation and coordination of the robot. Sensors including gyroscope and accelerometer modules were interfaced with the microcontroller to monitor orientation and maintain balance during walking. The microcontroller was programmed to execute walking sequences and coordinate joint movements for smooth locomotion.

Different motion patterns such as forward movement, backward movement, and turning actions were successfully developed and tested. Balance control techniques were implemented to improve the robot's stability during motion and reduce the possibility of falling. The mechanical and electronic components were fabricated, assembled, and integrated into a complete working prototype. Extensive testing and calibration of the robot were conducted on different surfaces to evaluate walking performance, stability, and adaptability.

Further analysis was carried out to study the robot's stability, walking speed, and energy consumption under different operating conditions. Motion control algorithms were continuously optimized through troubleshooting and performance improvements to achieve smoother and more reliable movement. Finally, the robot demonstrated both autonomous and controlled bipedal locomotion successfully. The project also included the study of real-world applications of bipedal robots in areas such as healthcare assistance, industrial automation, rescue missions, and hazardous environment operations.

1.4 CHAPTER ORGANIZATION

This report is structured into seven chapters:

- **Chapter 1** introduces the project, explains the background, problem statement, scope, and objectives.
- **Chapter 2** reviews related research and literature surveys.
- **Chapter 3** presents the design approach, system block diagram, methodology, and hardware implementation involved in the development of the proposed project.
- **Chapter 4** presents the experimental results and outcomes.
- **Chapter 5** concludes the work and suggests future improvements.
- **Chapter 6** contains the certification in international conference.
- **Chapter 7** references.

CHAPTER 2

LITERATURE REVIEW

The development of bipedal locomotion robots has become an important area of research in the field of robotics and automation due to their ability to imitate human walking and perform tasks in environments designed for humans. Researchers have focused on improving robot stability, walking efficiency, balance control, and autonomous movement using advanced sensors, actuators, and intelligent control systems. Earlier studies mainly concentrated on achieving stable walking through simple mechanical structures and static balancing techniques. With the advancement of embedded systems, artificial intelligence, and motion control algorithms, modern bipedal robots are capable of performing dynamic walking, obstacle avoidance, and adaptive locomotion on uneven terrains. This project focuses on the design and fabrication of a bipedal locomotion robot capable of stable movement, balance maintenance, and efficient control using suitable mechanical and electronic systems.

2.1 HUMANOID ROBOTS

S. Kajita, et. al., 2003 - Humanoid robots are designed to resemble the human body with components such as head, torso, arms, and legs. Researchers developed humanoid robots to perform tasks in human environments using human-like walking patterns and motion control techniques. These robots use servo motors, gyroscopes, accelerometers, and advanced controllers to maintain balance and coordination during movement. Humanoid robots are capable of performing activities such as walking, climbing stairs, object handling, and communication with humans.

Humanoid robots are widely used in healthcare assistance, research laboratories, education, and service industries. The development of humanoid robots has shown that proper integration of mechanical design, sensor feedback, and intelligent control systems is essential for achieving stable and smooth

bipedal locomotion. Researchers continue to improve humanoid robots by developing lightweight structures, efficient actuators, and intelligent navigation systems for better human interaction and mobility. [1]

2.2 STATIC AND DYNAMIC WALKING

M. Vukobratovic, et. al., 2004 - Early bipedal robots mainly used static walking mechanisms where the centre of gravity was maintained within the support area of the feet during movement. Static walking robots provide high stability but move slowly and consume more time during locomotion. These robots are suitable for applications where safety and balance are more important than speed.

Recent research focuses on dynamic walking robots which maintain balance while continuously moving using momentum and real-time feedback systems. Dynamic walking robots imitate natural human walking and achieve faster locomotion using advanced balancing techniques. Researchers have developed several gait generation methods to improve walking smoothness and reduce vibration during movement. Dynamic walking systems require accurate synchronization between sensors, actuators, and control algorithms. Various experiments show that dynamic locomotion improves mobility and enables robots to walk on uneven terrains, climb stairs, and recover from disturbances effectively. [2]

2.3 MICROCONTROLLER-BASED CONTROL

M. Hirose, et. al., 2007 - Microcontroller-based control systems play a major role in the development of low-cost and efficient bipedal robots. Controllers such as Arduino, Raspberry Pi, and PIC microcontrollers are widely used for controlling servo motors, processing sensor data, and executing walking algorithms.

The use of embedded systems allows real-time processing of movement instructions and synchronization between robot joints, improving overall locomotion performance and stability. [3]

2.4 SENSOR-BASED BALANCE CONTROL

M. H. Raibert 1996 - Maintaining balance is one of the major challenges in bipedal locomotion robots. Various studies have focused on the use of sensors such as gyroscopes, accelerometers, ultrasonic sensors, and force sensors to monitor robot orientation and body posture during movement.

Sensor fusion techniques are widely used to combine data from multiple sensors for accurate balance control and obstacle detection. Real-time sensor feedback improves walking accuracy and reduces instability. [4]

2.5 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

J. Pratt, et. al., 2006 - Artificial intelligence and machine learning techniques are increasingly used in modern bipedal robots to improve autonomous decision-making and adaptive locomotion. AI-based algorithms help robots analyze environmental conditions and modify walking patterns according to terrain conditions. Intelligent systems can process sensor data and make decisions regarding balance correction, obstacle avoidance, and navigation.

Machine learning techniques improve movement efficiency through continuous training and performance optimization. Intelligent control systems also help reduce energy consumption and improve coordination between joints. Researchers are exploring neural networks and deep learning techniques to develop robots capable of learning from their environment and adapting to different walking conditions. [5]

2.6 SERVO MOTOR CONTROL

B. Siciliano, et. al., 2016 - Servo motors are commonly used in bipedal robots because they provide precise angular movement and accurate position control. Researchers have implemented multiple servo motors in robot joints to achieve coordinated movement similar to human walking.

The use of servo motors improves walking flexibility, turning capability, and posture control. Proper synchronization of servo motors is essential for smooth and stable locomotion. [6]

2.7 INVERSE KINEMATICS

J. J. Craig, 2005 - Inverse kinematics is an important concept used in bipedal robotics for calculating joint angles required to achieve desired leg movement. Researchers use inverse kinematic models to improve walking accuracy and reduce unnecessary motion. These calculations help determine the position and orientation of robot joints during movement.

The implementation of inverse kinematics helps robots achieve efficient gait patterns, better balance, and coordinated movement between joints during locomotion. Researchers have also integrated inverse kinematics with trajectory planning algorithms to produce smooth and stable walking patterns similar to human locomotion. The use of inverse kinematics reduces mechanical stress on joints and improves the overall efficiency of the robot. It also enables precise control of foot placement while walking on complex surfaces. [7]

2.8 ZERO MOMENT POINT (ZMP)

M. W. Spong, et. al., 2006 - Zero Moment Point (ZMP) is one of the widely used balancing techniques in dynamic bipedal robots. The ZMP method helps maintain robot stability by ensuring that the resultant moment acting on the robot remains within the support area.

Researchers have used ZMP-based control algorithms to improve walking stability during motion. This method is highly effective for dynamic walking and humanoid robot applications. [8]

2.9 PATH PLANNING TECHNIQUES

K. S. Fu, et. al., 1987 - Path planning techniques are used in bipedal robots to determine safe and efficient movement paths. Researchers have developed algorithms that help robots avoid obstacles and navigate complex environments.

Advanced path planning methods improve autonomous navigation and reduce collision risks. These techniques are widely used in service robots, rescue robots, and industrial robotic systems. [9]

2.10 WIRELESS CONTROL SYSTEMS

J. M. Hollerbach, 1982 - Wireless control systems are used in many bipedal robots for remote operation and monitoring. Technologies such as Bluetooth, Wi-Fi, and RF communication are commonly used for transmitting control signals.

Wireless systems improve user convenience and allow operators to control robot movement from a distance. These systems are mainly used in research, surveillance, and experimental applications. [10]

2.11 LIGHTWEIGHT MATERIALS

J. Yamaguchi, et. al., 1999 - Researchers have focused on using lightweight materials such as aluminum alloys, acrylic sheets, and composite materials for robot fabrication. Lightweight structures reduce energy consumption and improve robot mobility. Material selection is very important because excessive weight affects balance, speed, and actuator performance.

The use of lightweight materials also reduces the load on actuators and enhances overall walking efficiency. Material selection plays an important role in robot durability and stability. Researchers have also explored the use of carbon fiber and 3D-printed materials to improve structural strength while maintaining low weight. Modern fabrication techniques allow the development of compact and durable robot frames with improved flexibility and shock absorption. These advancements help improve robot stability during walking and reduce mechanical wear. [11]

2.12 AUTONOMOUS NAVIGATION

J. H. Park, et. al., 1998 - Autonomous navigation enables bipedal robots to move independently without human intervention. Researchers have developed autonomous systems using sensors, cameras, and intelligent algorithms for environmental analysis.

Autonomous navigation improves robot adaptability and enables robots to perform tasks in dynamic environments. This technology is widely used in advanced humanoid and rescue robots. [12]

2.13 OBSTACLE DETECTION SYSTEMS

E. R. Westervelt, et, al., 2003 - Obstacle detection systems are important for preventing collisions during robot movement. Ultrasonic sensors, infrared sensors, and vision systems are commonly used for obstacle identification.

Researchers have shown that real-time obstacle detection improves robot safety and enhances navigation performance in indoor and outdoor environments. [13]

2.14 ENERGY EFFICIENCY TECHNIQUES

S. Collins, et. al., 2005 - Energy efficiency is an important research area in bipedal robotics because walking robots consume considerable power during movement. Researchers have developed optimized gait patterns and efficient motor control techniques to reduce energy usage. Power-efficient walking methods improve battery life and increase operational time.

Improved energy efficiency increases operating time and enhances robot performance. Efficient power management is especially important for battery-operated humanoid robots. Researchers are also exploring regenerative systems and lightweight actuators to reduce overall power consumption. Advanced energy management techniques help robots perform long-duration operations without frequent charging. Efficient power utilization also improves the reliability and practicality of bipedal robots in real-world applications. [14]

2.15 APPLICATIONS OF BIPEDAL ROBOTS

A. Goswami et. al., 1999 - Bipedal robots are widely used in healthcare, industrial automation, defense, rescue operations, and research applications. In healthcare, these robots assist elderly and disabled individuals in rehabilitation and mobility support. They are also used in hospitals for patient assistance and service activities.

Researchers are also exploring the use of bipedal robots in disaster management and space exploration where human-like movement is advantageous. The ability of bipedal robots to navigate stairs and uneven terrains makes them highly useful in challenging environments. In industrial sectors, bipedal robots can perform hazardous and repetitive tasks, reducing human effort and improving workplace safety. Future developments in bipedal robotics are expected to focus on advanced artificial intelligence, improved balance systems, and energy-efficient locomotion techniques. The increasing demand for intelligent automation systems has encouraged researchers to design robots capable of operating independently in complex and dynamic environments. [15]

CHAPTER 3

METHODOLOGY

3.1 GENERAL INTRODUCTION

Robotics is one of the fastest-growing fields of technology that combines mechanical engineering, electronics, computer science, and artificial intelligence to develop intelligent machines capable of performing human tasks. Among various robotic systems, bipedal locomotion robots have gained significant importance because of their ability to walk and move using two legs similar to human beings. These robots are specially designed to imitate human locomotion and operate effectively in environments created for humans such as homes, hospitals, industries, and public places. The study of bipedal robots plays an important role in improving automation, mobility, and intelligent robotic applications.

Bipedal locomotion is a complex process because maintaining balance during movement requires continuous coordination between mechanical structures, sensors, actuators, and control systems. Unlike wheeled robots, bipedal robots can walk on uneven surfaces, climb stairs, cross obstacles, and move in narrow spaces. To achieve stable locomotion, researchers use advanced technologies such as servo motor control, gyroscope sensors, accelerometers, inverse kinematics, and Zero Moment Point (ZMP) techniques. These technologies help the robot maintain balance and perform smooth walking movements.

The design and fabrication of a bipedal locomotion robot involve the integration of mechanical components, embedded systems, and intelligent control algorithms. The mechanical structure must be lightweight, strong, and flexible enough to support stable movement. Electronic systems including microcontrollers, sensors, and actuators are used to control joint movement and

monitor robot orientation. Proper synchronization between hardware and software is essential for achieving coordinated locomotion and balance control.

Bipedal robots have a wide range of applications in modern society. In healthcare, they assist elderly and disabled individuals during rehabilitation and mobility support. In industrial sectors, they can perform repetitive and hazardous tasks that reduce human effort and improve workplace safety. These robots are also useful in disaster management, military operations, space exploration, and research laboratories where human-like movement is required. The ability of bipedal robots to adapt to different environments makes them highly valuable for future intelligent automation systems.

The main objective of this project is to design and fabricate a stable and efficient bipedal locomotion robot capable of walking and maintaining balance. This project provides practical knowledge in robotics, embedded systems, motion control, and mechanical design. It also helps in understanding the challenges involved in human-like locomotion and intelligent robotic systems. With continuous advancements in artificial intelligence and automation technologies, bipedal robots are expected to play a major role in future robotic applications and smart environments.

3.2 PROCESS OF HARDWARE

The block diagram represents the overall hardware architecture of the bipedal robot control system. The complete system is powered using a 7.4V rechargeable battery, which acts as the main power source for the robot. Since different electronic components require regulated voltage levels for safe operation, the battery output is connected to a buck converter. The buck converter reduces and regulates the voltage to suitable levels required by the controller, servo driver, and servo motors.

The ESP32 microcontroller functions as the main controller of the robot. It is responsible for generating motion commands, controlling walking sequences,

and coordinating the movement of both legs. The ESP32 communicates with the PCA9685 PWM driver module through the I2C communication protocol. This communication method reduces the number of control pins required and allows efficient control of multiple servo motors simultaneously.

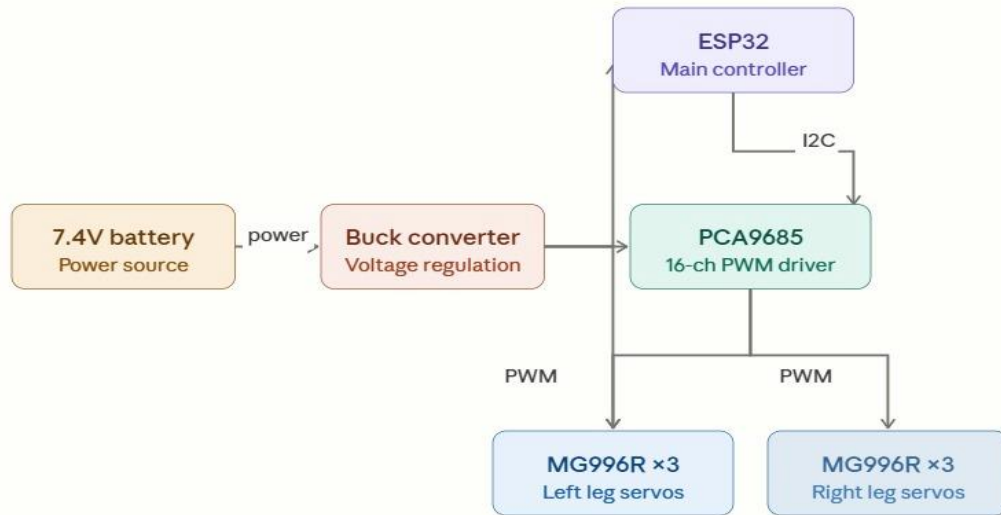


Fig 3.1: Proposed Process of Hardware

The PCA9685 module is a 16-channel PWM driver used to generate accurate PWM (Pulse Width Modulation) signals required for controlling the servo motors. The PWM signals produced by the driver are sent to the servo motors connected to the robot legs. In this modified system, each leg contains two MG996R Servo Motor servo motors instead of three. Therefore, a total of four servo motors are used in the robot, where two servos control the joints of the left leg and two servos control the joints of the right leg.

The servo motors provide the mechanical movement necessary for walking, balancing, and turning operations. The left-leg servos and right-leg servos work together in synchronization to imitate human-like walking motion. The ESP32 continuously processes the programmed walking algorithm and sends commands to the PCA9685 driver, which then controls the angular position of each servo motor precisely. This coordinated operation enables the robot to

perform stable forward, backward, and turning movements while maintaining balance during locomotion.

3.3 WORKING OPERATION

The working operation of the bipedal robot begins with the 7.4V battery supplying electrical power to the entire system. The battery output is connected to a buck converter, which regulates the voltage to suitable levels required for the electronic components and servo motors. This regulated power is then distributed to the ESP32 controller, PCA9685 PWM driver, and the servo motors.

The ESP32 acts as the main processing and control unit of the robot. It stores the walking program and continuously executes motion control algorithms for locomotion and balance. Based on the programmed instructions, the ESP32 generates movement commands for the robot such as forward walking, backward walking, and turning motions.

The ESP32 communicates with the PCA9685 PWM driver using the I2C communication protocol. The PCA9685 module receives the control data from the ESP32 and generates accurate PWM (Pulse Width Modulation) signals for each servo motor. These PWM signals determine the angular position and movement speed of the servo motors.

The robot uses four MG996R Servo Motor servo motors, with two motors placed in each leg. One servo motor controls the hip joint movement while the other controls the knee joint movement. During operation, the servo motors move in a coordinated sequence to imitate human walking patterns. When one leg moves forward, the other leg supports the robot body and maintains balance. The synchronized movement of both legs enables stable bipedal locomotion.

The walking cycle is achieved by continuously changing the angular positions of the servo motors according to the programmed gait sequence. The ESP32 sends updated position commands to the PCA9685 driver in real time,

ensuring smooth and controlled movement. The robot can also perform directional changes by varying the motion pattern of the left and right leg servos.

Overall, the integrated operation of the power supply system, ESP32 controller, PCA9685 PWM driver, and servo motors allows the robot to perform stable and controlled two-legged movement similar to human walking.

3.4 SELECTION OF MATERIALS

The materials used in the design and fabrication of a bipedal locomotion robot are selected carefully because the overall performance, stability, efficiency, and durability of the robot depend greatly on the quality and properties of the materials used. A bipedal robot continuously experiences dynamic loads, vibrations, joint movements, and balancing forces during walking. Therefore, the materials selected for different parts of the robot must possess suitable mechanical and electrical properties to ensure smooth locomotion and reliable operation. Important factors considered during material selection include strength, weight, flexibility, durability, cost, corrosion resistance, availability, and ease of manufacturing.

Strength is one of the most important factors because the robot structure must support the total weight of the body, motors, sensors, and electronic components without deformation or failure. At the same time, the material should be lightweight because excessive weight increases the load on actuators and affects balance and movement efficiency. Lightweight materials reduce power consumption, improve walking speed, and enhance robot stability. Materials such as aluminium alloys, acrylic sheets, and composite materials are commonly preferred because they provide a good balance between strength and low weight.

Flexibility and durability are also important considerations in bipedal robot design. Since robot joints undergo continuous movement and repeated stress during walking, the materials must withstand fatigue and mechanical wear for long operating periods. Flexible materials help absorb vibrations and shocks

generated during locomotion, reducing the possibility of structural damage. Durable materials improve the lifespan of the robot and reduce maintenance requirements. In advanced robotic systems, materials like carbon fiber composites are used because they provide excellent strength, flexibility, and durability while maintaining low weight.

Cost and ease of fabrication are additional factors that influence material selection, especially in prototype and educational projects. Materials that are easy to cut, drill, shape, and assemble simplify the fabrication process and reduce manufacturing time. Acrylic sheets and plastic materials are often used in small-scale robots because they are economical and easy to machine. Aluminium is also widely used because it provides good machinability and corrosion resistance. Proper selection of low-cost yet efficient materials help in developing reliable robotic systems within a limited budget.

Different types of materials and components are used for various sections of the robot including the structural frame, joints, actuators, sensors, electronic circuits, and power supply systems. Structural materials form the body and support system of the robot, while actuators such as servo motors provide controlled movement to the joints. Sensors such as gyroscopes and accelerometers are selected for balance control and orientation detection. Electronic components including microcontrollers, PCBs, and connecting wires are selected for control and communication purposes. Rechargeable batteries and voltage regulators are used to supply stable electrical power to the robot. Proper material selection directly affects robot stability, movement efficiency, energy consumption, safety, and operational life. A well-designed bipedal locomotion robot requires an effective combination of lightweight structural materials, efficient actuators, reliable sensors, and durable electronic systems. Therefore, material selection is considered one of the most important stages in the successful design and fabrication of a bipedal robotic system.

3.4.1 STRUCTURAL MATERIALS

Structural materials are the primary materials used for constructing the body, legs, joints, and supporting frame of a bipedal locomotion robot. These materials form the mechanical structure that supports all other components such as motors, sensors, batteries, and control systems. Since a bipedal robot continuously performs walking, balancing, and turning movements, the structural materials must possess high strength, low weight, durability, and resistance to mechanical stress. Proper selection of structural materials improves robot stability, movement efficiency, shock absorption, and overall operational life. The materials must also be easy to fabricate, machine, and assemble for effective robot construction.

3.4.1.1 ALUMINIUM

Aluminium is one of the most widely used structural materials in robotic systems because of its excellent combination of lightweight and high strength. It has a high strength-to-weight ratio, which makes it suitable for bipedal robots where reducing body weight is very important for maintaining balance and reducing power consumption. Aluminium is also corrosion-resistant, which protects the robot structure from environmental damage and increases durability. Due to its low density, aluminium reduces the load on actuators and servo motors, allowing smoother and more efficient locomotion.



Fig. 3.2: Aluminium Frame

Another advantage of aluminium is its excellent machinability. It can be easily cut, drilled, shaped, welded, and assembled into different robot components. Aluminium frames provide sufficient rigidity to support robot joints and moving parts without excessive deformation during walking. Because of these advantages, aluminium alloys are commonly used in robot legs, joint supports, body frames, and structural brackets in both prototype and industrial robotic systems.

3.5 HARDWARE ENVIRONMENT

3.5.1 LM2596 BUCK CONVERTER

The LM2596 Buck Converter Module is a DC-DC step-down voltage regulator used to reduce and regulate the input voltage from the battery to a lower stable output voltage required by the electronic components of the bipedal robot. In this project, the robot is powered using a 7.4V battery, but components such as the ESP32 controller and servo driver require regulated voltage for safe and stable operation. The LM2596 buck converter performs this voltage regulation efficiently.

The LM2596 module works on the principle of switching regulation, where the input voltage is rapidly switched on and off and then filtered to produce a smooth and lower DC output voltage. Compared to linear regulators, the buck converter provides high efficiency with less heat generation and lower power loss, making it suitable for battery-powered robotic systems.

In the bipedal robot, the 7.4V battery output is connected to the input terminals of the LM2596 buck converter. The converter then reduces the voltage to the desired level, such as 5V or 6V, depending on the requirements of the ESP32 controller, PCA9685 PWM driver, and the MG996R Servo Motor servo motors. The regulated output ensures stable operation of all electronic components and prevents damage caused by voltage fluctuations.



Fig. 3.3: LM2596 Buck Converter

The LM2596 module also includes an adjustable potentiometer that allows the output voltage to be set accurately. It provides protection features such as thermal shutdown, current limiting, and short-circuit protection, improving the reliability and safety of the robotic system.

3.5.2 BATTERY 7.4V

The 7.4V Lithium-Ion Battery is used as the primary power source for the bipedal locomotion robot. It provides the electrical energy required for operating all the components of the system, including the ESP32 controller, PCA9685 PWM driver, sensors, and the MG996R Servo Motor servo motors. The battery ensures continuous and stable power delivery during the walking and balancing operations of the robot.

The 7.4V battery is typically made by connecting two 3.7V lithium-ion cells in series. This arrangement provides sufficient voltage and current capacity needed for driving multiple servo motors simultaneously. Lithium-ion batteries are commonly used in robotic systems because they are lightweight, compact, rechargeable, and capable of storing a large amount of energy compared to their size.



Fig. 3.4: 7.4V Battery

In the robot system, the battery output is connected to the LM2596 buck converter, which regulates the voltage to suitable levels required by different electronic components. The battery supplies high current to the servo motors during movement and also powers the control circuits responsible for robot operation. Since the servo motors consume higher current during walking and turning motions, the battery must provide stable voltage without excessive power loss.

The use of a 7.4V lithium-ion battery improves the portability and mobility of the robot by allowing wireless operation without external power connections. Its compact size and efficient energy storage make it suitable for mobile robotic applications where lightweight construction and longer operating time are important. Thus, the battery plays a vital role in ensuring reliable performance, efficient power management, and smooth operation of the bipedal locomotion robot.

3.5.3 ESP32

The ESP32 is a powerful and low-cost microcontroller widely used in robotics, automation, embedded systems, and Internet of Things (IoT) applications. It is developed by Espressif Systems and is known for its high processing speed, compact size, and integrated wireless communication capabilities. In the bipedal locomotion robot, the ESP32 acts as the main control

unit responsible for controlling the overall operation of the robot. It processes programmed instructions, generates walking sequences, and coordinates the movement of the servo motors to achieve stable bipedal locomotion. The controller continuously executes motion control algorithms and ensures proper synchronization between the left and right legs during walking and turning operations.



Fig. 3.5: ESP32

One of the major advantages of the ESP32 is its ability to perform multiple tasks simultaneously due to its dual-core processing capability. It supports real-time control applications and provides fast response during robotic movement and sensor data processing. The ESP32 also consumes relatively low power, making it suitable for battery-operated robotic systems. In addition to controlling servo motors, the ESP32 can interface with sensors such as gyroscopes and accelerometers to improve balance and stability. Its built-in Wi-Fi and Bluetooth features also allow wireless communication and future expansion for remote monitoring or smartphone-based robot control.

The architecture of the ESP32 consists of a dual-core Tensilica Xtensa LX6 microprocessor operating at a clock speed of up to 240 MHz. The dual-core architecture allows one core to handle communication tasks while the other manages real-time control operations. The controller includes built-in Wi-Fi and

Bluetooth modules, SRAM memory, flash memory support, timers, watchdog circuits, ADCs, DACs, PWM generators, and several communication interfaces such as UART, SPI, I2C, and CAN. These integrated peripherals reduce the need for additional external hardware and make the ESP32 highly efficient for embedded system applications.

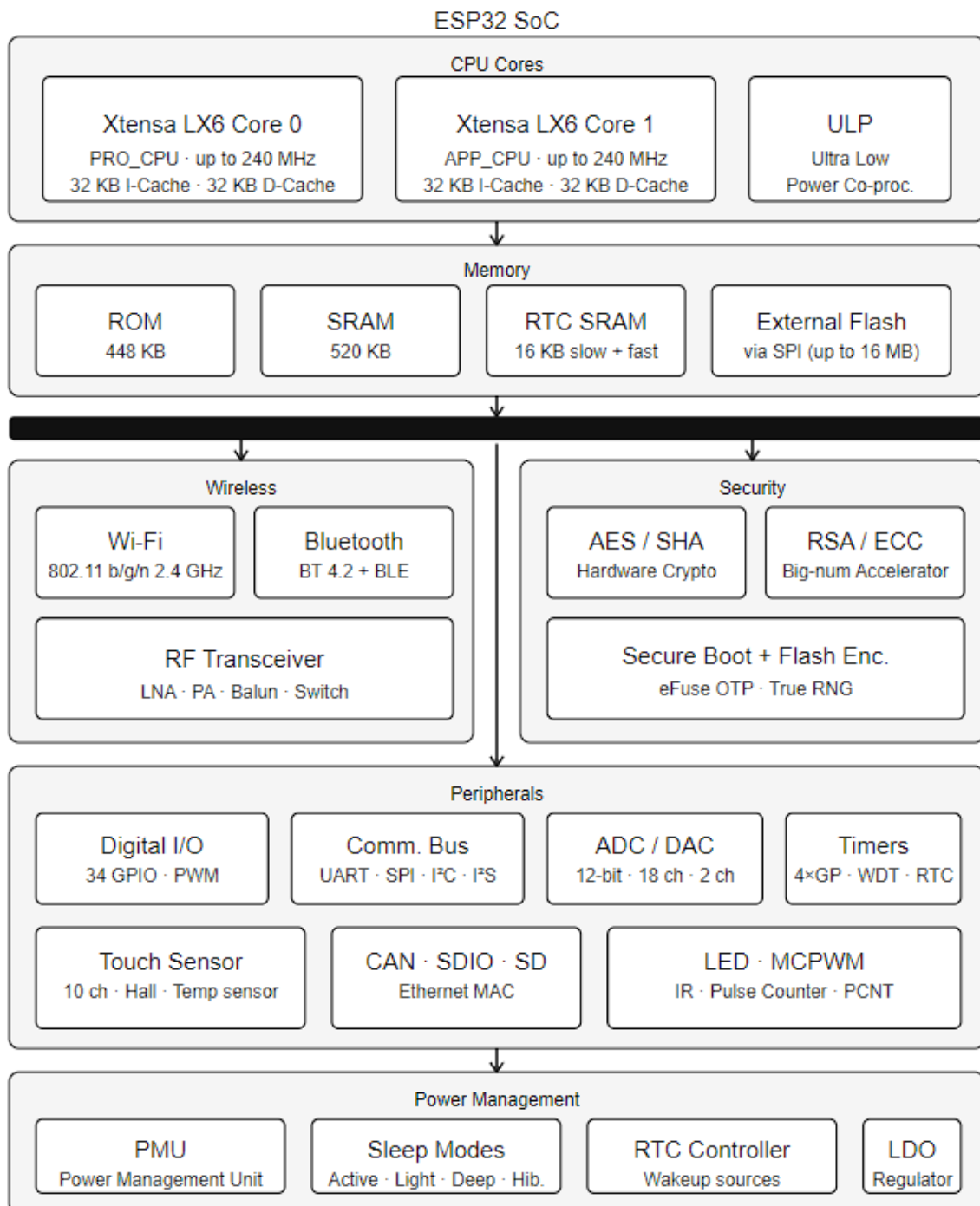


Fig. 3.6: ESP32 Architecture

The ESP32 architecture is designed for high-speed data processing and efficient multitasking. It supports low-power operating modes, which help in reducing energy consumption during standby conditions. The microcontroller also includes hardware acceleration features for cryptography and secure communication. In the bipedal robot system, the ESP32 communicates with the PCA9685 PWM driver through the I2C protocol to generate precise servo control signals. The processing capability of the ESP32 enables smooth execution of motion control programs, balance correction algorithms, and sensor interfacing simultaneously without affecting system performance.

Table 3.1: Pin Configuration of ESP32

Pin Number	Pin Name	Description
1	VIN / 5V	Input power supply pin used to provide operating voltage to the ESP32 board
2	GND	Ground pin used for common electrical reference
3	EN	Enable pin used to reset or enable the ESP32 controller
4	GPIO 21 (SDA)	I2C data pin used for communication with the PCA9685 servo driver
5	GPIO 22 (SCL)	I2C clock pin used for communication with the PCA9685 servo driver
6	GPIO 1 (TX)	UART transmit pin used for serial communication
7	GPIO 3 (RX)	UART receive pin used for serial communication
8	GPIO 34	Analog input pin used for sensor interfacing
9	GPIO 35	General-purpose input pin for sensor connections
10	3.3V	Regulated 3.3V output pin for external modules and sensors

The ESP32 provides multiple General Purpose Input Output (GPIO) pins that can be configured for various functions depending on the application requirements. In the bipedal locomotion robot, GPIO 21 and GPIO 22 are commonly used as SDA and SCL pins respectively for I2C communication with the PCA9685 servo driver module. Power pins such as VIN and GND are used to provide the required operating voltage and grounding connection to the controller. The ESP32 also includes ADC pins for analog sensor interfacing and UART pins for serial communication and programming purposes. The flexible pin configuration allows easy interfacing with different sensors, actuators, and external modules.

The pin configuration of the ESP32 is highly versatile because many GPIO pins support multiple functionalities such as PWM generation, SPI communication, UART communication, ADC input, DAC output, and touch sensing. In this project, the PWM signals required for controlling the MG996R Servo Motor servo motors are generated indirectly through the PCA9685 module controlled by the ESP32. Additional GPIO pins can also be connected to sensors such as gyroscope and accelerometer modules for balance monitoring and feedback control. The availability of numerous configurable pins makes the ESP32 highly suitable for robotic applications requiring multiple sensor and actuator connections.

3.5.3.1 PROGRAM

```
#include <Wire.h>
#include <Adafruit_PWMServoDriver.h>

Adafruit_PWMServoDriver pca = Adafruit_PWMServoDriver(0x40);

// PULSE RANGE

int minPulse = 70;
```

```

int maxPulse = 720;

int toPulse(int angle) {

    return map(angle, 0, 180, minPulse, maxPulse);

}

#define CH_KNEE_LEFT  5

#define CH_KNEE_RIGHT 6

#define CH_HIP_LEFT   7

#define CH_HIP_RIGHT

// KNEE_MOVE = 20 (was 10)

// HIP_MOVE = 30 (was 20)

#define CENTER    90

#define KNEE_MOVE  20 // was 10 — increased for more lift

#define HIP_MOVE   30 // was 20 — increased for bigger step

// LEFT LEG — goes above center

#define L_KNEE_BENT    (CENTER + KNEE_MOVE) // 110

#define L_KNEE_STRAIGHT CENTER              // 90

#define L_HIP_FORWARD  (CENTER + HIP_MOVE) // 120

#define L_HIP_CENTER   CENTER              // 90

// RIGHT LEG — goes below center (FLIPPED to fix backward
movement)

#define R_KNEE_BENT    (CENTER - KNEE_MOVE) // 70

#define R_KNEE_STRAIGHT CENTER              // 90

```

```

#define R_HIP_FORWARD  (CENTER - HIP_MOVE)  // 60

#define R_HIP_CENTER    CENTER                // 90

// SPEED SETTINGS

// STEP_DELAY reduced from 50 to 30 = faster movement

// PAUSE    reduced from 400 to 250 = less gap between actions

#define STEP_DELAY  30  // was 50 — increased speed

#define PAUSE      250  // was 400 — less gap between actions

// POSITION TRACKER

int pos[16];

// CORE: sendAngle

void sendAngle(int channel, int angle) {

    if (angle < 0)  angle = 0;

    if (angle > 180) angle = 180;

    pca.setPWM(channel, 0, toPulse(angle));

}

// CORE: moveOne — smooth single servo movement

void moveOne(int channel, int target) {

    if (target < 0)  target = 0;

    if (target > 180) target = 180;

    int current = pos[channel];

    Serial.print("  Ch"); Serial.print(channel);

    Serial.print(" "); Serial.print(current);

```

```

Serial.print(" -> "); Serial.print(target);

Serial.println(" deg");

if (current < target) {

    for (int a = current; a <= target; a++) {

        sendAngle(channel, a);

        pos[channel] = a;

        delay(STEP_DELAY);

    }

} else {

    for (int a = current; a >= target; a--) {

        sendAngle(channel, a);

        pos[channel] = a;

        delay(STEP_DELAY);

    }

}

sendAngle(channel, target);

pos[channel] = target;

delay(PAUSE);

}

// CORE: moveTogether — two servos move at same time smoothly

void moveTogether(int ch1, int t1, int ch2, int t2) {

    Serial.print("  Ch"); Serial.print(ch1);

```

```

Serial.print(" -> "); Serial.print(t1);

Serial.print(" + Ch"); Serial.print(ch2);

Serial.print(" -> "); Serial.print(t2);

Serial.println(" (together)");

int f1 = pos[ch1];

int f2 = pos[ch2];

int steps = max(abs(t1 - f1), abs(t2 - f2));

if (steps == 0) return;

for (int s = 0; s <= steps; s++) {

    int a1 = f1 + (t1 - f1) * s / steps;

    int a2 = f2 + (t2 - f2) * s / steps;

    sendAngle(ch1, a1);

    sendAngle(ch2, a2);

    pos[ch1] = a1;

    pos[ch2] = a2;

    delay(STEP_DELAY);

}

sendAngle(ch1, t1);

sendAngle(ch2, t2);

pos[ch1] = t1;

pos[ch2] = t2;

```

```

    delay(PAUSE);

}

// LEFT LEG — knee flex + hip flex simultaneously

void leftSwingPhase() {

    Serial.println("--- LEFT: swing ---");

    moveTogether(

        CH_KNEE_LEFT, L_KNEE_BENT,

        CH_HIP_LEFT, L_HIP_FORWARD

    );

}

void leftStancePhase() {

    Serial.println("--- LEFT: stance ---");

    moveTogether(

        CH_KNEE_LEFT, L_KNEE_STRAIGHT,

        CH_HIP_LEFT, L_HIP_CENTER

    );

}

// RIGHT LEG — flipped angles fix backward movement

void rightSwingPhase() {

    Serial.println("--- RIGHT: swing ---");

    moveTogether(

        CH_KNEE_RIGHT, R_KNEE_BENT,

```



```

    CH_HIP_RIGHT, R_HIP_FORWARD

);

}

void rightStancePhase() {

    Serial.println("--- RIGHT: stance ---");

    moveTogether(

        CH_KNEE_RIGHT, R_KNEE_STRAIGHT,

        CH_HIP_RIGHT, R_HIP_CENTER

    );

}

// FULL WALK CYCLE

void walkForward() {

    Serial.println("===== WALK =====");

    Serial.println("-- LEFT --");

    leftSwingPhase();

    leftStancePhase();

    Serial.println("-- RIGHT --");

    rightSwingPhase();

    rightStancePhase();

    Serial.println("===== DONE =====");

}

// SETUP

```

```

void setup() {

  Serial.begin(115200);

  Wire.begin(21, 22);

  pca.begin();

  pca.setPWMFreq(50);

  Serial.println("=====");

  Serial.println(" ROBOT STARTING");

  Serial.println(" Ch5 = Knee LEFT");

  Serial.println(" Ch6 = Knee RIGHT (flipped)");

  Serial.println(" Ch7 = Hip LEFT");

  Serial.println(" Ch8 = Hip RIGHT (flipped)");

  Serial.println("=====");

  Serial.println(" KNEE_MOVE = 20 deg");

  Serial.println(" HIP_MOVE = 30 deg");

  Serial.println(" STEP_DELAY = 30ms");

  Serial.println("=====");

  for (int i = 0; i < 16; i++) {

    pos[i] = CENTER;

  }

  Serial.println("Walking now...");

  delay(500);

}

```

```
// LOOP

void loop() {

    walkForward();

    delay(200);

}
```

3.5.4 PCA9685 SERVO DRIVER

The PCA9685 Servo Driver Module is a PWM (Pulse Width Modulation) controller module used for controlling multiple servo motors efficiently using I2C communication. It is widely used in robotics and automation systems where accurate and synchronized control of several servo motors is required. In the bipedal locomotion robot, the PCA9685 module is used to control the four MG996R Servo Motor servo motors responsible for leg movement and walking operations. The module helps reduce the processing load on the ESP32 by independently generating precise PWM signals for each servo motor.

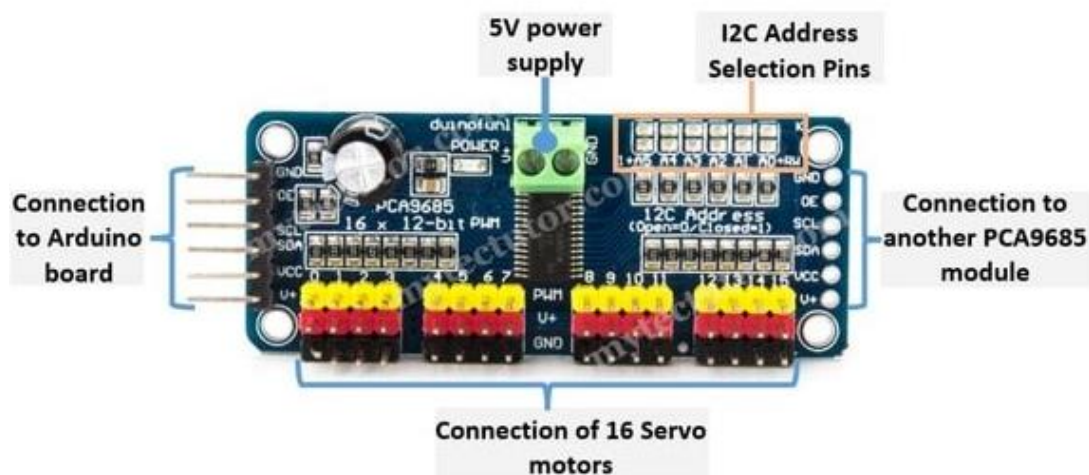


Fig. 3.7: PCA9685 Servo Driver

The PCA9685 servo driver operates through the I2C communication protocol and is interfaced with the ESP32 controller using the SDA and SCL communication pins. The ESP32 sends movement commands and position data

to the PCA9685 module, which then converts these commands into PWM signals required for controlling the angular movement of the servo motors. This arrangement allows smooth, stable, and synchronized movement of the robot joints during walking, balancing, and turning actions. Since the module can control up to sixteen servo motors simultaneously, it provides flexibility for future expansion of the robotic system.

The PCA9685 module contains several important pins for communication, power supply, and servo control operations. The VCC and GND pins are used to provide power and grounding for the logic circuit of the module. The SDA and SCL pins are used for I2C communication with the ESP32 controller. The V+ terminal is used to supply external power to the servo motors because servo motors require higher current during operation. The module also contains multiple PWM output channels where the servo motors are connected. An Output Enable (OE) pin is provided to enable or disable PWM signal output when required. These pin connections allow efficient communication and accurate control of multiple servo motors in the bipedal locomotion robot system.

3.5.5 MG996R SERVO MOTOR

The MG996R Servo Motor is a high-torque servo motor widely used in robotics, automation systems, and motion control applications. It is specially designed to provide precise angular movement and strong rotational force, making it suitable for robotic joints and locomotion mechanisms. In the bipedal locomotion robot, the MG996R servo motors are used to control the movement of the robot legs and joints. These motors play a major role in generating walking motion, maintaining balance, and performing directional movements such as forward walking, backward walking, and turning operations.

The MG996R servo motor operates using Pulse Width Modulation (PWM) control signals generated by the PCA9685 Servo Driver Module and controlled by the ESP32. Based on the PWM signal width, the servo motor rotates to a

specific angular position with high accuracy. The motor contains an internal DC motor, gearbox, control circuit, and feedback mechanism that continuously monitors the shaft position and corrects any positional error automatically. This closed-loop control mechanism allows smooth and accurate movement of the robot joints during locomotion.



Fig. 3.8: MG996R Servo Motor

In this project, four MG996R servo motors are used, with two motors connected to each leg of the robot. One servo motor controls the hip joint movement while the other controls the knee joint movement. The synchronized movement of these servo motors enables the robot to imitate human walking patterns. The servo motors provide sufficient torque to support the robot body weight and maintain stable movement during operation. Their fast response, durability, and high torque capability make them highly suitable for bipedal robotic applications.

The MG996R servo motor consists of three main connection wires used for operation and control. The VCC wire is used for supplying operating voltage to the motor, the GND wire provides the ground connection, and the signal wire receives PWM control signals from the PCA9685 servo driver. The motor generally operates at voltages between 4.8V and 7.2V and can rotate

approximately 180 degrees depending on the control signal. The compact design, metal gear mechanism, and high torque output of the MG996R servo motor help improve the overall stability, efficiency, and walking performance of the bipedal locomotion robot.

3.6 SOFTWARE ENVIRONMENT

3.5.1 Arduino Software (IDE)

The Arduino Software Integrated Development Environment (IDE) is a powerful and user-friendly platform used to write, compile, and upload programs to Arduino and compatible microcontroller boards. It provides a simple programming interface based on C and C++, allowing beginners and professionals to develop embedded applications with ease. The IDE includes a built-in text editor, compiler, and serial monitor, making it a complete development environment for hardware programming.



Fig. 3.9: Arduino IDE

One of the main strengths of the Arduino IDE is its simplicity and accessibility. Users can write programs called "sketches" and upload them to a connected board with a single click. The IDE automatically handles complex processes such as compiling code, managing device drivers, and communicating with the hardware. This makes Arduino ideal for learning electronics, robotics, automation, and Internet of Things (IoT) applications. One of the main strengths of the Arduino IDE is its simplicity and accessibility. Users can write programs

called "sketches" and upload them to a connected board with a single click. The IDE automatically handles complex processes such as compiling code, managing device drivers, and communicating with the hardware. This makes Arduino ideal for learning electronics, robotics, automation, and Internet of Things (IoT) applications.

Additionally, the Arduino IDE supports a wide variety of microcontroller platforms through the Board Manager. It allows developers to work not only with Arduino boards but also with ESP32, NodeMCU, STM32, and many other third-party boards. With the help of a vast library ecosystem and online community support, Arduino IDE has become one of the most widely used platforms in embedded system development.

Step 1: Open a web browser and visit the official Arduino website (<https://www.arduino.cc>).

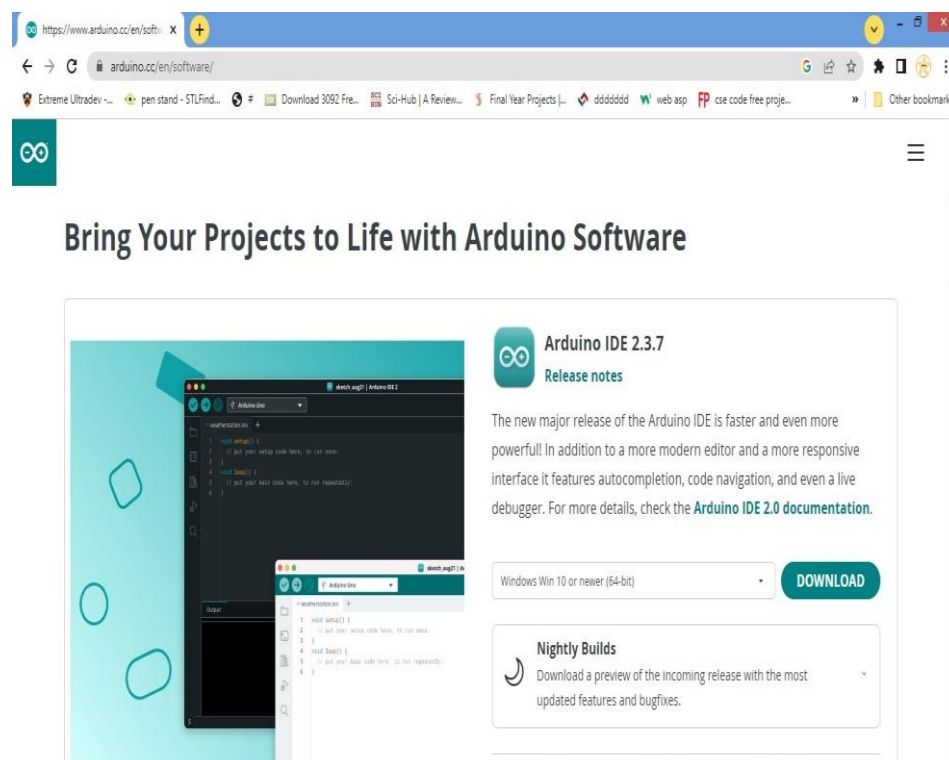


Fig. 3.10: Official Arduino Website

Step 2: Click on the Software or Download option and choose Arduino IDE for Windows.

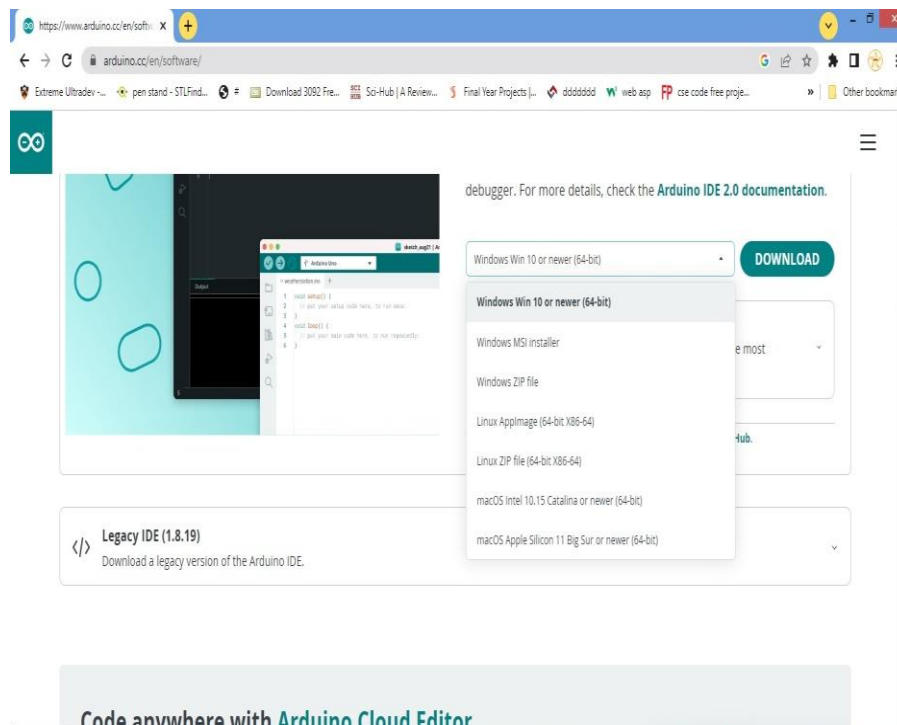


Fig. 3.11: Arduino IDE for Windows

Step 3: Download the installer (.exe file).

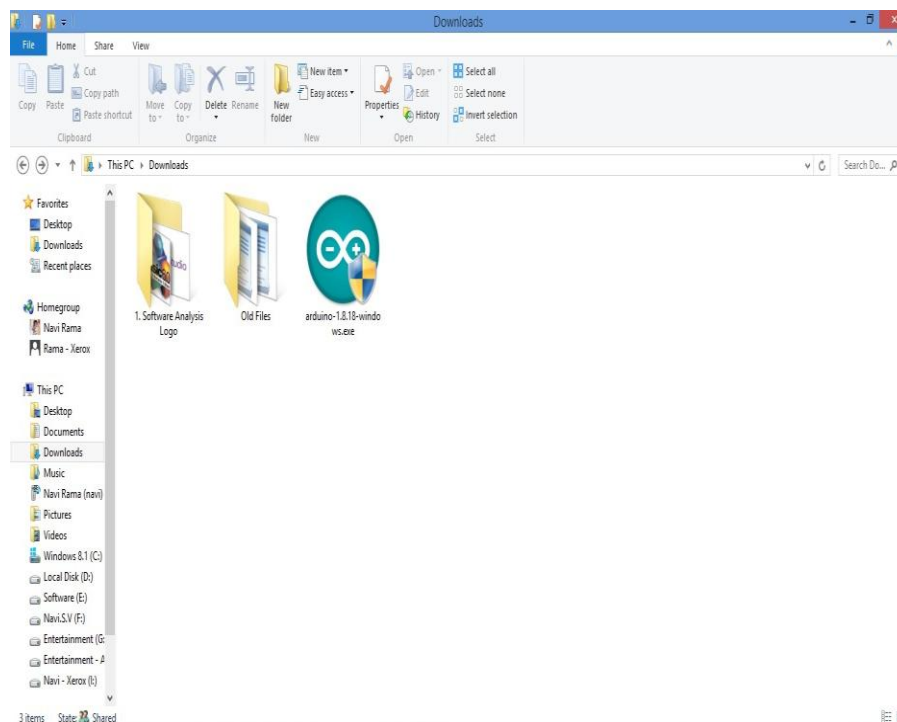


Fig. 3.12: Download the installer File

Step 4: After the download is complete, double-click the installer file.

Step 5: Accept the license agreement and choose the installation components.

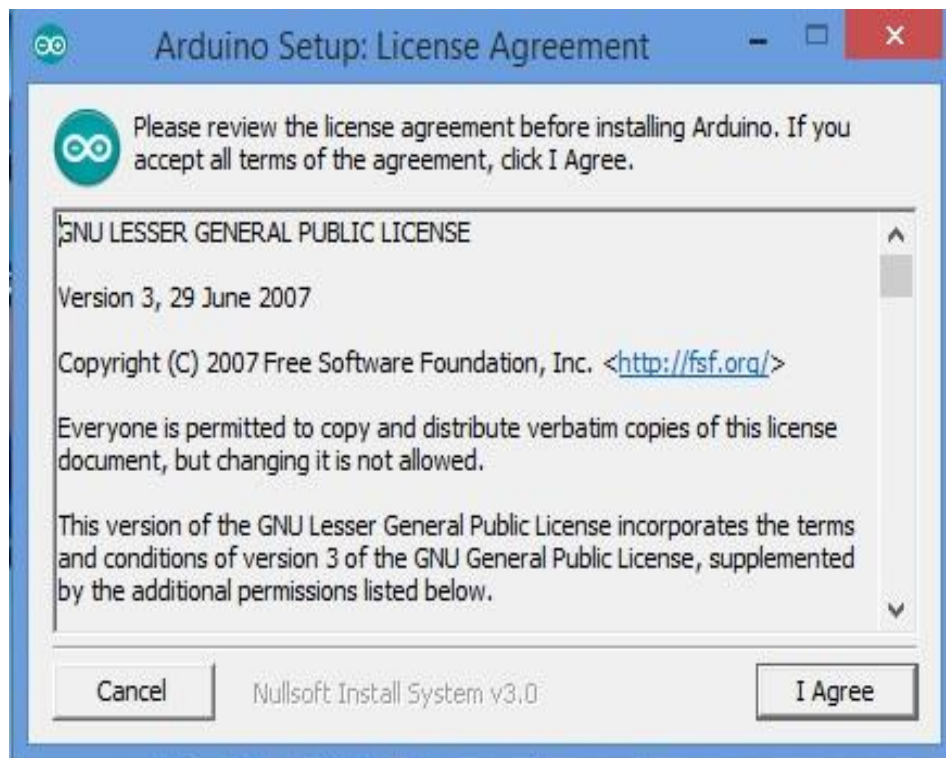


Fig. 3.13: Accept the license agreement

Step 6: Select the installation folder and click Install.

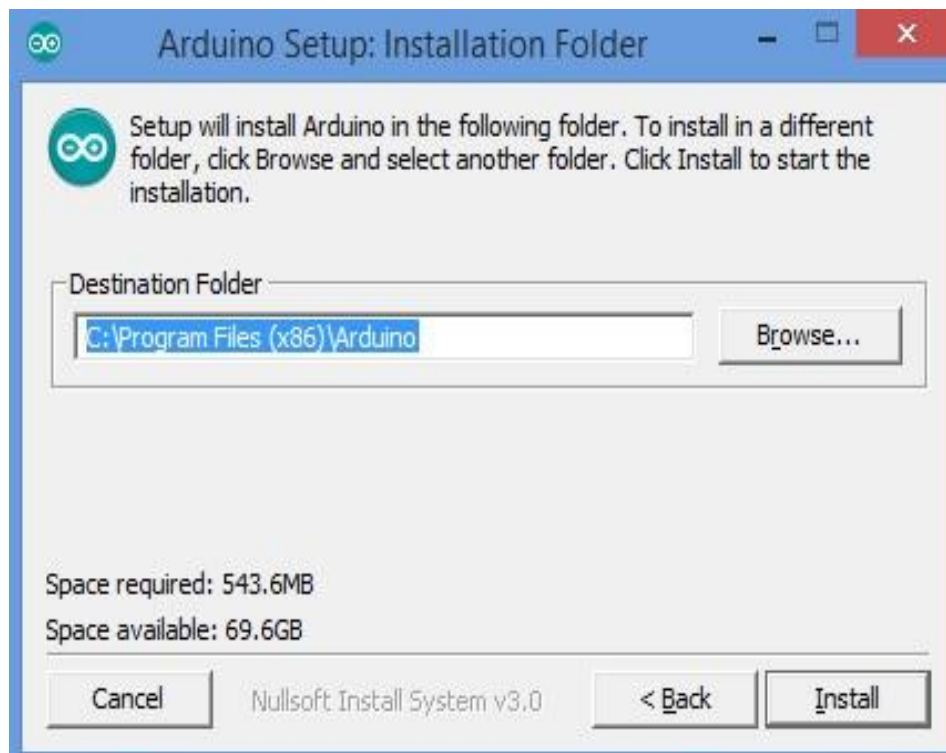


Fig. 3.14: Installation Folder

Step 7: Allow Windows to install the required USB drivers when prompted.

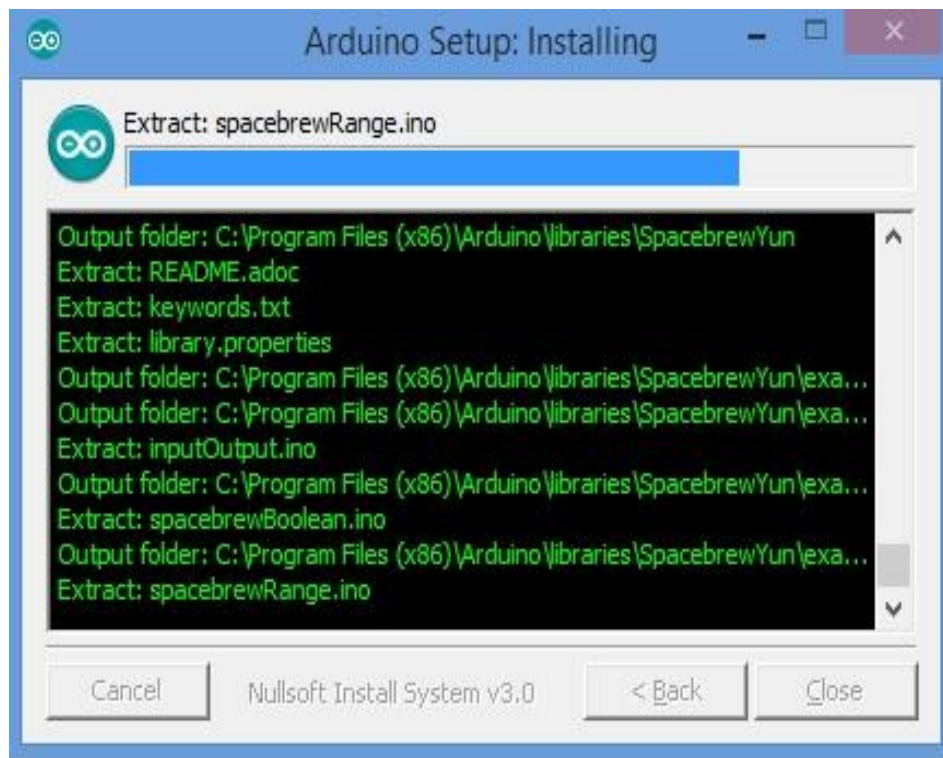


Fig. 3.15: Install USB Driver

Step 8: After installation, click Finish to complete the setup.

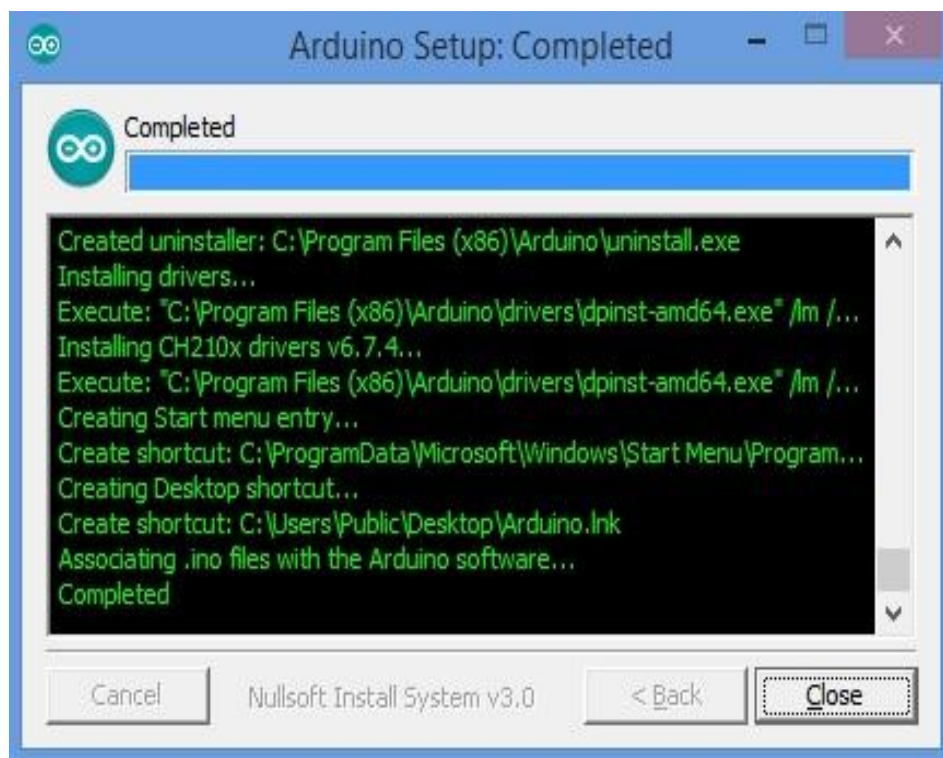


Fig. 3.16: Complete the Setup

Step 9: Launch the Arduino IDE from the desktop or Start menu.



Fig. 3.17: Launch the Arduino IDE

Step 10: Connect your Arduino board to the computer using a USB cable.

Arduino supports a wide range of microcontroller boards through its Board Manager, which allows users to install and manage board definitions easily. Popular Arduino boards such as Arduino Uno, Mega, and Nano are based on ATmega microcontrollers. NodeMCU boards use the ESP8266 microcontroller, which offers built-in Wi-Fi for IoT applications. ESP32 boards provide dual-core processing, Bluetooth, and Wi-Fi, making them suitable for advanced wireless and real-time projects. STM32 boards are based on ARM Cortex microcontrollers and are widely used for high-performance and industrial applications. By using the Arduino Board Manager, users can install support packages for all these boards and program them using the same Arduino IDE.

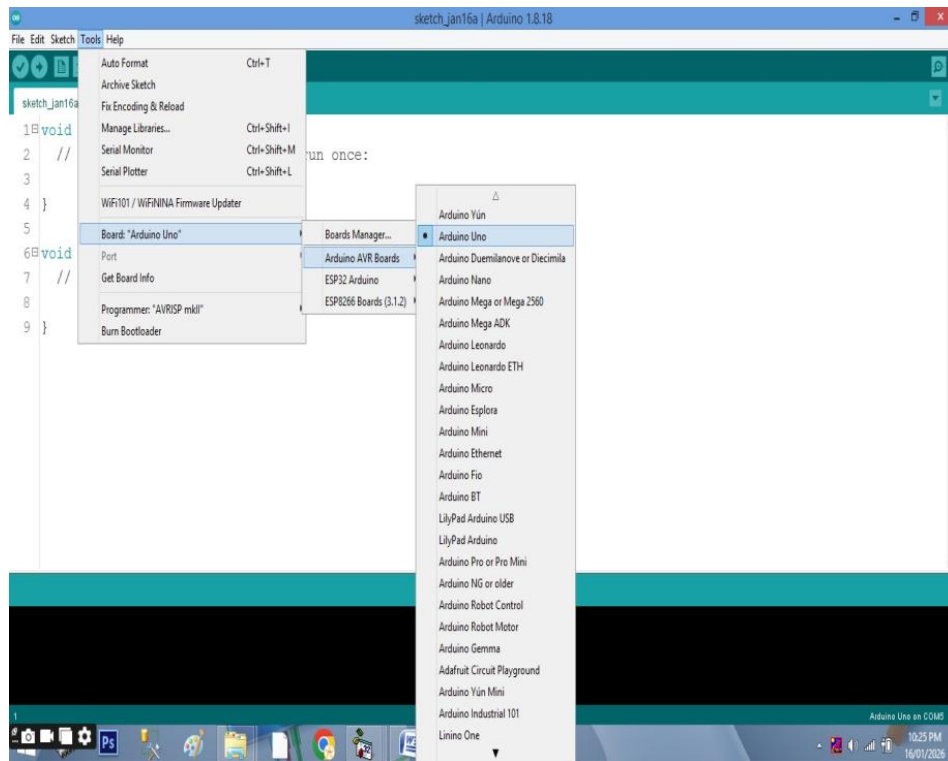


Fig. 3.18: Board Manager

Libraries in Arduino are collections of pre-written code that make it easier to use sensors, displays, motors, and communication modules. To install a library, open the Arduino IDE and go to Sketch ? Include Library ? Manage Libraries. The Library Manager window will open, displaying thousands of available libraries. In the search bar, type the name of the library you need, such as WiFi, Servo, or LCD. When the required library appears, click the Install button.

The IDE will automatically download and add the library to your system. Once installed, the library can be included in your program using the `#include` command. Using libraries reduces development time and improves code reliability because they are tested and optimized by the Arduino community. Library management allows users to update, remove, or reinstall libraries whenever required, making Arduino development flexible and efficient for both beginners and advanced users.

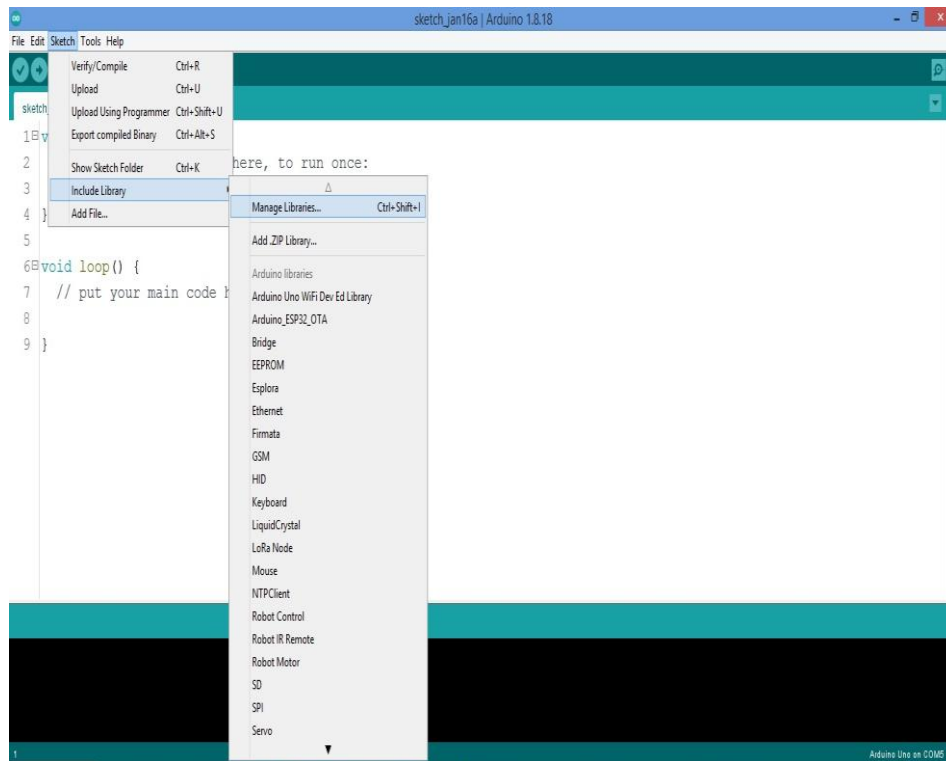


Fig. 3.19: Library Management

The Arduino Software IDE (Integrated Development Environment) is the primary platform used to write, compile, and upload embedded programs to Arduino microcontroller boards. It provides a simple and user-friendly interface where users can write code in a language based on C and C++, specifically optimized for embedded systems. The Arduino IDE includes a code editor, compiler, library manager, and serial monitor, all in one package. This allows developers to create programs (called sketches) that interact with hardware components such as sensors, motors, LEDs, displays, and communication modules. In embedded coding with Arduino, the program structure is very simple and consists of two main functions: `setup()` and `loop()`. The `setup()` function runs only once when the board is powered on or reset, and it is used to initialize pins, communication protocols, and variables.

The `loop()` function runs repeatedly in an infinite cycle, making it ideal for continuously reading sensor values, checking conditions, and controlling outputs.

For example, if a temperature sensor is connected, the Arduino continuously reads its value in the loop and controls a fan or alarm based on the reading. This looping behavior is what makes Arduino suitable for real-time embedded applications.

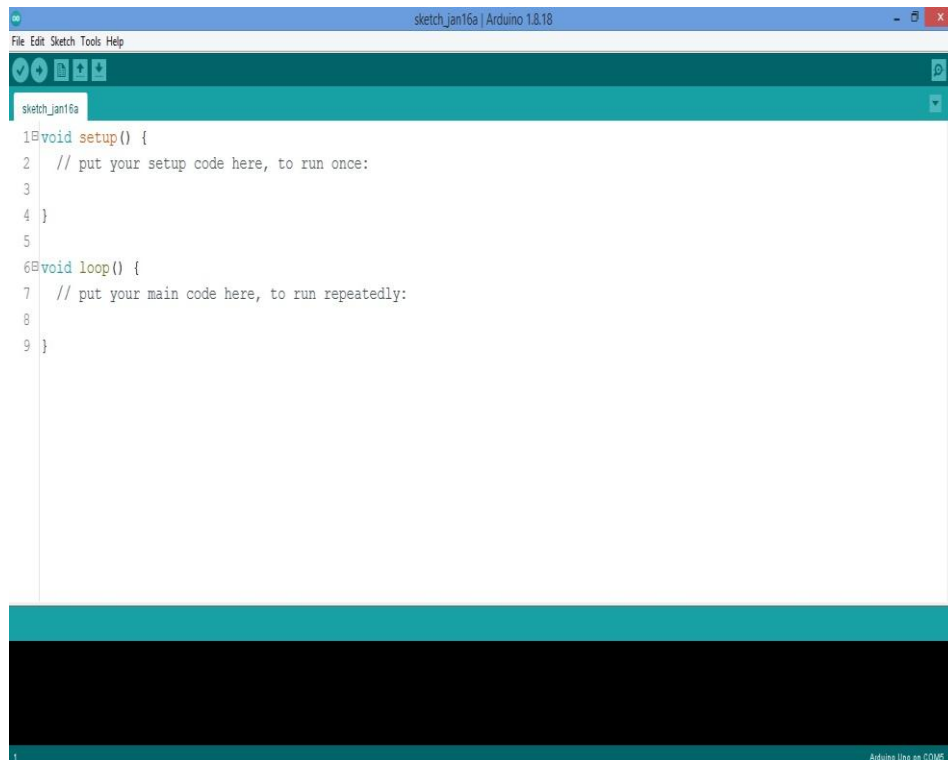


Fig. 3.20: Arduino Code Editor

The Arduino IDE also provides a large collection of built-in and external libraries, which simplify complex tasks such as communication (I2C, SPI, UART), Wi-Fi, Bluetooth, and LCD control. These libraries allow developers to use simple function calls instead of writing low-level hardware control code. Once the program is written, the IDE compiles it and uploads it to the microcontroller using a USB cable. The Serial Monitor in the IDE helps in debugging by displaying real-time data from the Arduino, making it easier to test and troubleshoot embedded systems. Overall, the Arduino Software IDE makes embedded system programming simple, powerful, and accessible for beginners and professionals alike.

CHAPTER 4

RESULT AND DISCUSSIONS

4.1 RESULT

The bipedal locomotion robot was successfully designed, fabricated, assembled, and tested for performing basic two-legged walking operations. The developed robotic system effectively integrated mechanical, electronic, and programming components to achieve controlled locomotion. The robot was powered using a 7.4V lithium-ion battery, while the LM2596 buck converter provided regulated voltage supply for the electronic circuits and servo motors.

The ESP32 microcontroller successfully executed the programmed walking sequences and controlled the movement of the robot through the PCA9685 servo driver module. The MG996R servo motors generated sufficient torque for performing walking movements and maintaining basic posture control.

During testing, the robot successfully performed forward walking, backward movement, and directional turning operations. The servo motors responded accurately to the PWM signals generated by the ESP32 controller through the PCA9685 module using I2C communication. The coordinated motion of the servo motors enabled the robot to imitate simple human walking patterns. The robot demonstrated satisfactory response to programmed motion commands and maintained controlled locomotion on smooth and level surfaces for short durations.

The fabricated mechanical structure was compact and lightweight, which reduced the load on the servo motors and improved portability. The robot frame and joint assembly successfully supported all electronic and mechanical components. The project also demonstrated the successful implementation of basic robotic concepts such as gait generation, servo motor coordination, embedded programming, and robotic motion control. The developed system

proved that basic bipedal locomotion can be achieved using low-cost components and microcontroller-based control techniques.

4.2 DISCUSSION

Although the robot successfully performed basic locomotion, stability issues were observed during continuous walking operations. The robot structure was slightly unstable during dynamic motion because the center of gravity continuously shifted between the two legs during walking. Due to this instability, the robot required external support while moving to prevent falling. The balance control mechanism implemented in the project was basic and insufficient for achieving completely autonomous unsupported locomotion.

The instability mainly occurred because of limited degrees of freedom, lightweight frame structure, vibration generated by the servo motors, and limitations in the walking algorithm. Since only a limited number of servo motors were used, the robot had restricted flexibility for advanced posture correction and balance adjustment. Accurate synchronization between servo motors was necessary for maintaining stability, and small timing differences occasionally caused imbalance during movement.

Surface conditions also influenced the performance of the robot. The robot showed better stability on smooth and flat surfaces, while rough or uneven surfaces increased instability and reduced locomotion efficiency. The walking speed remained moderate because the robot required careful balance maintenance during operation.

Energy consumption also increased during continuous walking because the MG996R servo motors consumed higher current while supporting the body weight and maintaining posture.

Despite these limitations, the project successfully demonstrated the working principle of a bipedal locomotion robot and achieved the main objectives of robot design and movement control. The project provided practical knowledge in embedded systems, servo motor coordination, robotic locomotion, gait

programming, and mechanical fabrication. Future improvements such as additional degrees of freedom, gyroscope and accelerometer-based feedback systems, PID control algorithms, and AI-based balancing techniques can significantly improve robot stability, walking efficiency, and autonomous locomotion capability.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The bipedal locomotion robot project was successfully completed with the design, fabrication, and implementation of a two-legged robotic system capable of performing basic walking operations. The project combined mechanical design, embedded systems, electronic interfacing, and motion control techniques to achieve controlled bipedal locomotion. The robot was developed using lightweight structural materials, MG996R Servo Motor servo motors, the ESP32 microcontroller, and the PCA9685 Servo Driver Module for accurate servo motor control. The system successfully demonstrated coordinated leg movement for forward walking, backward motion, and directional turning.

The project provided practical understanding of the principles involved in human walking mechanisms, gait generation, balance control, and robotic joint coordination. The ESP32 controller successfully executed the programmed walking sequences and communicated effectively with the PCA9685 servo driver using I2C communication. The servo motors responded accurately to PWM signals and generated the required joint movements for locomotion. The integration of the battery power supply and LM2596 buck converter ensured stable operation of the entire robotic system.

During testing and experimentation, the robot demonstrated basic bipedal movement successfully on flat surfaces. The project achieved the primary objective of implementing a functional bipedal robotic prototype capable of simulating simple human walking patterns. The robot also provided valuable insights into robotic motion control, synchronization of actuators, structural design limitations, and embedded programming techniques. The project helped

improve technical skills related to robotics, electronics, microcontroller programming, circuit integration, and troubleshooting.

However, certain limitations were observed during the operation of the robot. The mechanical structure was slightly unstable during continuous movement because maintaining dynamic balance in a two-legged robot is a complex task. The robot required external support while walking to prevent falling due to the shifting center of gravity and limited balance correction capability. The use of only two servo motors in each leg reduced the degrees of freedom and restricted advanced movement flexibility. Despite these limitations, the project successfully demonstrated the working concept of a low-cost bipedal locomotion system and served as a strong foundation for future developments in humanoid robotics.

Overall, the project proved that bipedal locomotion can be achieved using simple mechanical design and microcontroller-based control systems. The developed robot successfully demonstrated the coordination between hardware and software components required for robotic walking. The project also highlighted the challenges associated with stability, synchronization, and real-time balance control in humanoid robots, thereby providing a valuable learning platform for further research and development in the field of robotics and automation.

5.2 FUTURE SCOPE

The future scope of the bipedal locomotion robot project is extensive and can be further enhanced by incorporating advanced technologies for intelligent movement, improved stability, and better human-robot interaction. One of the major future improvements involves implementing dynamic obstacle detection and autonomous navigation systems. Sensors such as ultrasonic sensors, infrared sensors, LiDAR modules, or camera-based vision systems can be integrated to detect obstacles and environmental changes in real time. This would allow the

robot to move safely in dynamic environments without human assistance and improve its capability for autonomous operation.

Advanced balance control mechanisms can also be implemented in future versions of the robot. Gyroscope and accelerometer sensors can be integrated with real-time feedback algorithms such as PID control, fuzzy logic, or machine learning-based control systems to achieve better posture correction and stability during walking. Additional servo motors can be added to increase the degrees of freedom in the leg joints, allowing smoother and more natural humanoid movement. Improved gait generation algorithms and dynamic walking techniques can help the robot achieve fully unsupported walking and enhanced mobility on uneven surfaces.

Another important future enhancement is the integration of voice control systems for intelligent interaction. Speech recognition modules and AI-based voice assistants can be used to allow the robot to respond to voice commands such as walking, stopping, turning, or changing direction. This feature can improve user interaction and make the robot more suitable for assistive and service-based applications. Wireless communication through Wi-Fi or Bluetooth can also be implemented for remote monitoring and smartphone-based control.

The project can further be extended by incorporating sign language recognition systems using computer vision and artificial intelligence techniques. A camera module and image processing algorithms can be used to detect and interpret hand gestures or sign language movements. The recognized sign language can then be converted into text and displayed on a screen attached to the robot. Similarly, voice commands can also be converted into text and displayed on the screen for better accessibility and communication support. This feature can be highly useful for assisting hearing-impaired and speech-impaired individuals in real-world applications.

Future developments may also include integrating AI-based decision-making systems for autonomous behavior and intelligent interaction. Machine learning algorithms can enable the robot to learn walking patterns, adapt to different surfaces, and improve movement efficiency over time. Cloud connectivity and IoT integration can also be implemented for remote data monitoring and smart robotic control. With these advancements, the bipedal locomotion robot can find applications in healthcare assistance, rehabilitation support, industrial automation, rescue operations, surveillance systems, and interactive humanoid robotics.

CHAPTER 6

CONFERENCE CERTIFICATIONS

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This is to certify that Mr./Ms./Dr. NAVANEETHAM K from		
DHIRAJLAL GANDHI COLLEGE OF TECHNOLOGY has presented a paper entitled		
BIPEDAL ASSISTIVE COMMUNICATION ROBOT WITH SIGN LANGUAGE DISPLAY AND VOICE INTERFACE		
in "International Conference on Sustainable Technologies and Advanced Materials (ICSTAM -26)" held on April 17 & 18, 2026 organized by the Department of Mechatronics, Mechanical, Agricultural & Civil Engineering, Bannari Amman Institute of Technology, Sathyamangalam - 638401, Erode, Tamil Nadu.		
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CHAPTER 7

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